
SWYNEX EDA Task 2

❖ Exploratory Data Analysis

This project is part of my internship task at SWYNEX Technologies.

The objective of this task is to perform Exploratory Data Analysis (EDA) on the cleaned Indian School Education Statistics dataset.

❖ Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Google Colab

❖ Dataset

The dataset contains Indian school education dropout statistics for different states/UTs and years.

The cleaned dataset from Task 1 was used for this analysis.

❖ ## Analysis Performed

The following analysis was performed:

1. Dataset size and structure
2. Missing value analysis
3. Duplicate record check
4. Descriptive statistics
5. Average dropout rate by education level
6. Yearly dropout rate analysis
7. Identification of unusually high dropout values
8. Data visualization using charts

❖ Visualizations

The analysis includes:

- Average dropout rate by education level
- Yearly average dropout trend
- Missing values by education level

❖ Key Insights

1. Secondary education has the highest average dropout rate.
2. Primary dropout rate decreased over the years.
3. Secondary dropout rate shows a slight increase over the years.
4. Some records have unusually high dropout values.
5. Higher Secondary has the highest number of missing values among the four total dropout columns.

❖ Files

- `Task2\_EDA.ipynb` – Google Colab notebook containing the complete analysis.
- `cleaned\_dropout\_ratio.csv` – Cleaned dataset used for the analysis.
- `README.md` – Project documentation.

❖ Conclusion

Exploratory Data Analysis helped identify important patterns, trends, missing values and unusual records in the Indian School Education Statistics dataset.

The analysis provides useful information about dropout rates across different education levels and years.